Hung Nguyen

hun004@ucsd.edu | (858)-241-1558 | La Jolla, CA, USA | Personal Page | LinkedIn | Google Scholar

EDUCATION

University of California San Diego, USA
M.S. and Ph.D., Electrical Engineering (Signal and Image Processing) 09/2023 – 06/2029 (expected)
- Advisors: Distinguished Profs. Truong Nguyen, Pamela Cosman.

Ho Chi Minh City University of Technology, Vietnam
B.Eng., Mechatronics Engineering 08/2017 – 06/2022

RESEARCH INTERESTS

- 3D/4D reconstruction, image/video/Gaussian splatting compression, generative AI, vision-language models.
- Frequency-domain transforms, e.g., wavelet, Fourier, for modern computer vision and robot learning architectures.

RESEARCH EXPERIENCE

Video Processing Lab, UC San Diego
Graduate Student Researcher 09/2024 – Present
- Research priors from frequency-domain transforms for sparse-view/compressed/radio-frequency Gaussian Splatting.

WorldQuant Vietnam
Quantitative Researcher Intern 01/2023 – 07/2023
- Research CNN architectures that transform time-series data into image representations for stock price prediction.

SELECTED PUBLICATIONS & PREPRINTS

First-authored publications
- **Hung Nguyen**, An Le, Runfa Li, Truong Nguyen. "From Coarse to Fine: Learnable Discrete Wavelet Transforms for Efficient 3D Gaussian Splatting", ICCV Workshops 2025. [Paper] [Poster]
- **Hung Nguyen**, Runfa Li, An Le, Truong Nguyen. "DWTGS: Rethinking Frequency Regularization for Sparse-view 3D Gaussian Splatting", VCIP 2025. [Paper] [Poster]
- **Hung Nguyen**, Runfa Li, An Le, Truong Nguyen. "WaveletGaussian: Wavelet-domain Diffusion for Sparse-view 3D Gaussian Object Reconstruction", ICASSP 2026. [Paper] [Poster]
- **Hung Nguyen**, An Le, Truong Nguyen. "Learnable Multi-level Discrete Wavelet Transforms for 3D Gaussian Splatting Frequency Modulation", EUSIPCO 2026. [Paper] [Poster]

Other publications
- An Le, **Hung Nguyen**, …, Truong Nguyen. "Stop-band Energy Constraint for Orthogonal Tunable Wavelet Units in Convolutional Neural Networks for Computer Vision problems", VCIP 2025 **Oral**. [Paper] [Talk]
- An D. Le, **Hung Nguyen**, …, Truong Q. Nguyen, Cheolhong An. "Universal Wavelet Units in 3D Retinal Layer Segmentation", ICAISF 2026 **Oral**. [Paper] [Talk]

ACADEMIC SERVICES

- Conference Reviewer: ICASSP (2027), WACV (2027), ISBI (2027).
- Journal Reviewer: TIP (2026), SPL (2026), OJSP (2026).

HONORS AND AWARDS

- Centaur Graduate Fellowship, ECE Department, UC San Diego. Awarded 02/2026.
- Center for Wireless Communications Fellowship, UC San Diego. Awarded 02/2026.
- IEEE Eta Kappa Nu Invitation for top 1/3 Graduate Students of UCSD ECE Department. Awarded 10/2024.
- Vingroup Scholarship for Overseas Study for Master's and Doctoral Degrees, Vingroup JSC. Awarded 06/2023.
- Panasonic Asia Scholarship for Undergraduate Students, Panasonic Vietnam. Awarded 04/2022.
- Scholarship for Undergraduate Engineering Students, PECC2 (Vietnam). Awarded 09/2021.

WORK EXPERIENCE

VinMotion, Vietnam

Senior AI Engineer 12/2025 – 05/2026

- Develop a data-efficient framework to transfer text-conditioned grasp detection to speech for humanoid robots.
- Develop a pipeline for humanoid robotic grasping based on segmentation and CAD-based pose estimation.

TECHNICAL SKILLS

- Programming Languages: Python, C/C++, MATLAB.
- DL/ML Packages: PyTorch, JAX, TensorFlow, Keras, standard image processing libraries (e.g., OpenCV, NumPy).
- DL/ML Methods: VLMs, Diffusion Models, Vision Transformers, NeRF/3DGS, CNNs.
- Robotics: ROS 2.